Computer Science Programs / Aldrin Torres 2026-2027

5 LONG-TERM PROJECTS

Program Theme

Digital Wellness, Human Behavior, and Technology

Computer Science Pathway

Digital Wellness Analytics Platform

Community Partner: Neuroscience & Psychology Club

Students develop a web-based analytics platform that investigates the relationship between technology usage habits and student wellness. Students create anonymous survey instruments, collect voluntary anonymous survey data, organize and analyze datasets, develop visualizations, and publish findings through an interactive website dashboard.

Implementation Status:

This project is scheduled to begin in December. Prior to implementation, students will collaborate with the Neuroscience and Psychology Club to refine research questions, identify appropriate wellness metrics, develop survey instruments, establish data collection procedures, and determine methods for sharing findings with the school community. The project plan will be finalized once consultation with the club sponsor and student leadership has been completed.

Deliverables

- Project Proposal
- Anonymous Survey Instrument
- Dataset

- Data Analysis Report
- Interactive Website Dashboard
- Final Presentation

Total Time

23 Hours

Industry Skills

- Data Collection
- Data Cleaning
- Data Visualization
- Front-End Web Development
- User Experience Design
- Software Testing
- Project Documentation

Georgia Standards Alignment

Introduction to Software Technology (IT-FND-4, IT-FND-5, IT-FND-6)

- Analyze and interpret data.
- Develop technology solutions to solve real-world problems.
- Apply web development and user interface design principles.
- Demonstrate project planning and documentation skills.
- Communicate technical findings to stakeholders.

AP Computer Science Principles

- Big Idea 1: Creative Development
- Big Idea 2: Data
- Big Idea 3: Algorithms and Programming
- Big Idea 5: Impact of Computing

Game Design Pathway

Digital Dopamine: Understanding Technology Through Interactive Play

Community Partner: Neuroscience & Psychology Club

Students develop a lightweight 2D Unity educational game that demonstrates how technology habits influence focus, attention, reward systems, habit formation, and digital wellness. The game emphasizes simple assets, scene transitions, educational gameplay mechanics, and accessibility within school hardware constraints.

Implementation Status:

This project is scheduled to begin in December following completion of foundational Unity and game development instruction. During the first semester, students are building prerequisite skills in game mechanics, programming logic, asset integration, project planning, and software development workflows. Students have not yet entered the primary development phase of the long-term project. The project timeline remains subject to adjustment based on instructional pacing and technology availability.

Technology access disruptions experienced during the semester resulted in adjustments to pacing for prerequisite skill development activities. As a result, the long-term capstone project will begin after students have completed the required preparation and foundational coursework.

Deliverables

- Game Design Document
- Storyboard
- Asset Library
- Playable Unity Prototype
- Testing Report
- Final Presentation

Total Time

22 Hours

Industry Skills

- Game Design
- Unity Development

- C# Programming
- User Interface Design
- Debugging
- Asset Management
- Interactive Storytelling

Georgia Standards Alignment

Game Design & Simulation Standards

- GDAS1: Explore careers and opportunities in game development.
- GDAS2: Apply principles of game design and development.
- GDAS3: Create digital assets and game components.
- GDAS4: Develop and test interactive games.
- GDAS5: Evaluate, revise, and present completed projects.

Computer Science Practices

- Computational Thinking
- Iterative Design
- Problem Solving
- Collaboration
- Software Testing

Blank Long-Term Project Rubric

Criteria	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)	Score
Project Planning					
Research & Documentation					
Technical Implementation					
Creativity & Innovation					
Team Collaboration					
Testing & Revision					

Communication & Presentation					
Professionalism					

Total Points: ____ / 32

Teacher Comments

Student Team List

Computer Science Pathway Teams

Team #	Team Name	Student 1	Student 2	Student 3	Student 4	Teacher Approval
1						
2						

Game Design Pathway Teams

Team #	Team Name	Student 1	Student 2	Student 3	Student 4	Teacher Approval
1						
2						

Project Approval Form

Team Name: _____

Project Title: _____

Pathway: ☐ Computer Science ☐ Game Design

Community Impact Statement:

Teacher/Consultant Approval: _____

Approval Date: _____

This addition makes Section 1.5 audit-ready by showing standards alignment, assessment methodology, student collaboration documentation, and approval procedures.